

Algebraic Topology: a beginner's course

This lecture:

**AlgTop25: More on the
fundamental group**

copyright 2010 N J Wildberger

Lecture 25: More on the fundamental group



$\pi(M, x_0)$ is actually a group.

(1) Identity: $e = [\alpha_0]$ constant loop

$$[\alpha_0] * [\alpha] = [\alpha_0 * \alpha] = [\alpha]$$

$$[\alpha] * [\alpha_0] = [\alpha]$$



$$\alpha_0 * \alpha \sim \alpha$$

so $[\alpha_0 * \alpha] = [\alpha]$



2. Inverses.

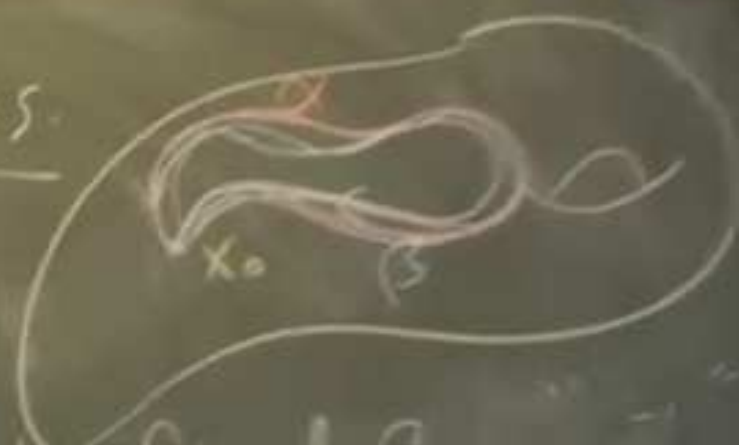
$$[\alpha] * [\beta] = e$$

Q Given α , how to find $\beta = \alpha^{-1}$

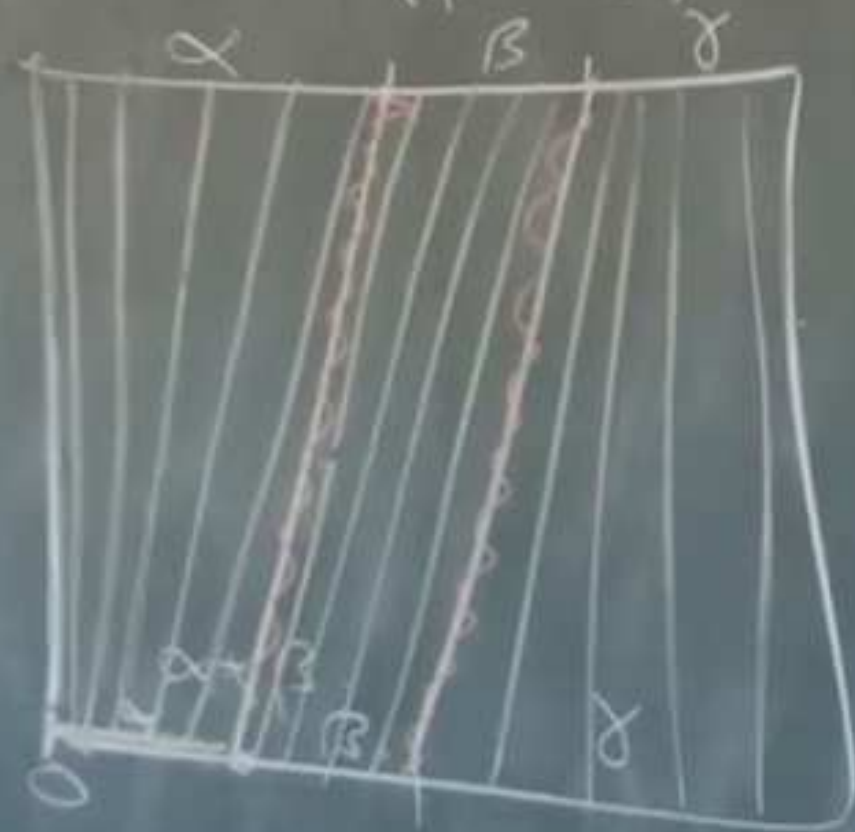
A Choose $\beta(t) \equiv \alpha(1-t)$

(Going back along α)

Show $\alpha * \beta \sim e$



3. Associativity: $\alpha * (\beta * \gamma) = (\alpha * \beta) * \gamma$

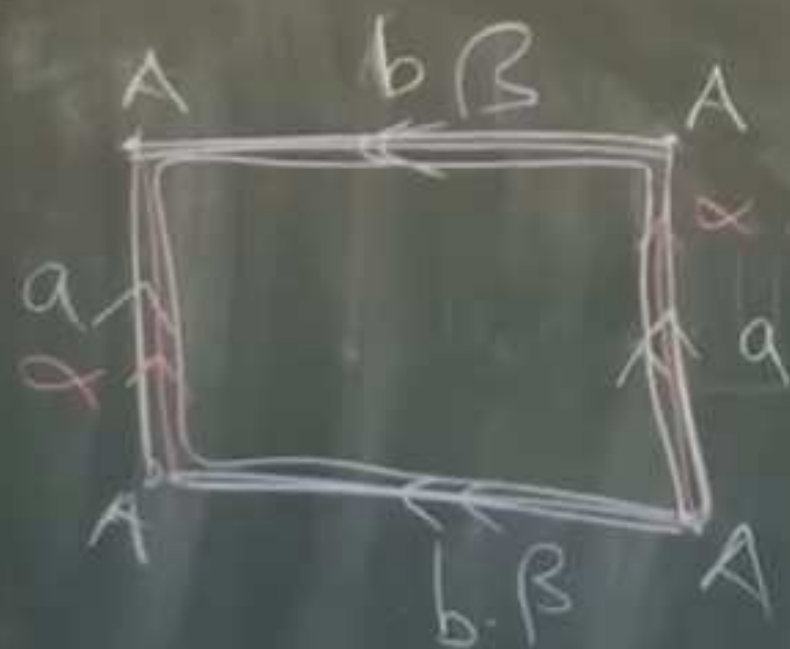


$(\alpha * \beta) * \gamma$





Two 'generators' of
 $\pi(T) : \alpha, \beta$



$aba^{-1}b^{-1}$

$$\alpha * \beta * \alpha^{-1} * \beta^{-1}$$



Two 'generators' of
 $\pi(T) : \alpha, \beta$



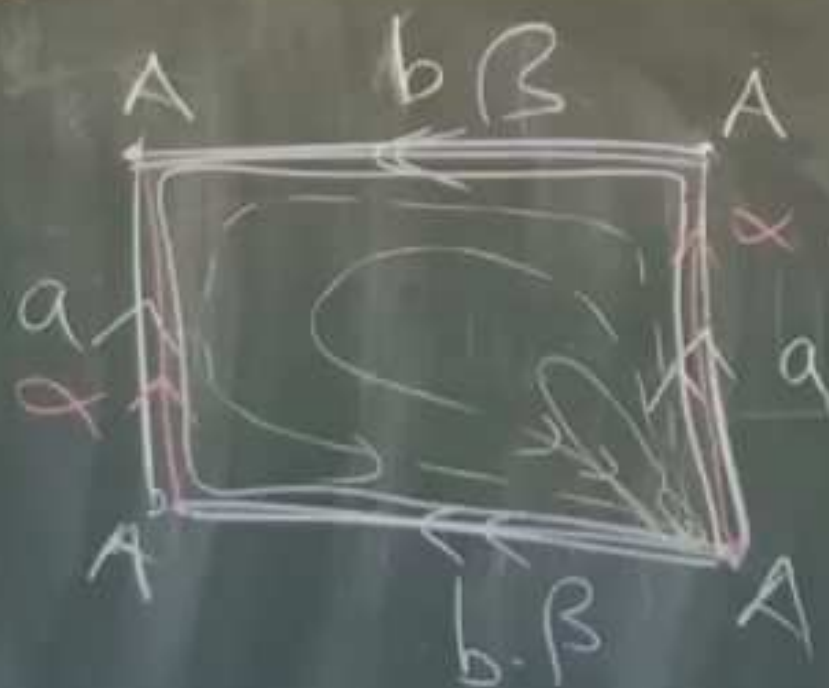
$$\alpha * \beta * \alpha^{-1} * \beta^{-1} \sim e$$

$$aba^{-1}b^{-1}$$



Two 'generators' of
 $\pi(T) : \alpha, \beta$

torus



$aba^{-1}b^{-1}$

$$\alpha * \beta * \alpha^{-1} * \beta^{-1} \sim e$$

$$\alpha \beta \alpha^{-1} \beta^{-1} = e$$

$$\Rightarrow \boxed{\alpha \beta = \beta \alpha}$$

Commuting



Two 'generators' of
 $\pi(T) : \alpha, \beta$

$$\alpha^2 \beta^3 \alpha^{-1} \beta = \alpha^2 \alpha^{-1} \beta^3 = \alpha \beta^3$$

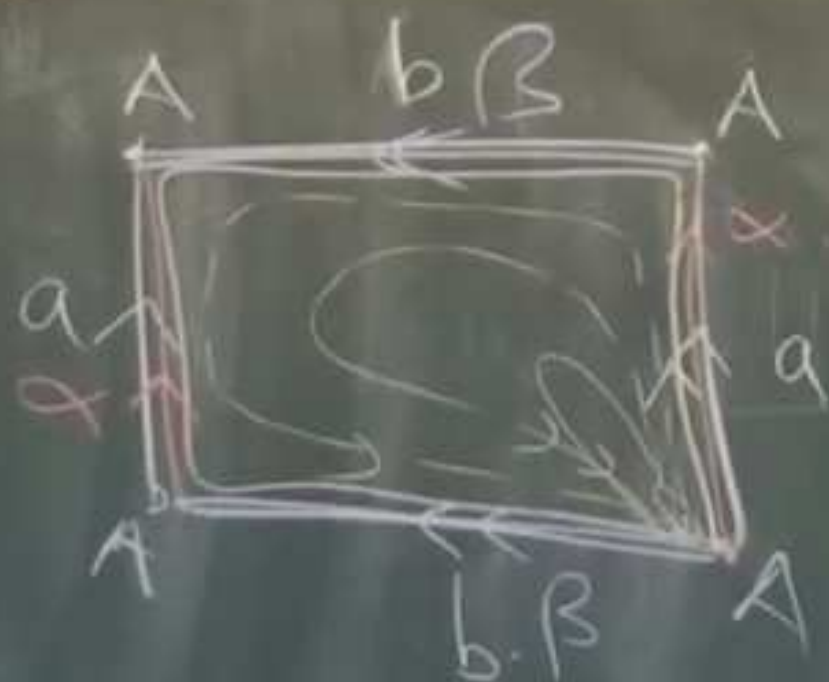
every el.

$\pi(T)$ a

$$\alpha^n \beta^m$$

$$\mathbb{Z} \times \mathbb{Z}$$

torus



$$a b a^{-1} b^{-1}$$

$$\alpha * \beta * \alpha^{-1} * \beta^{-1} \sim e$$

$$\alpha \beta \alpha^{-1} \beta^{-1} = e$$

$$\Rightarrow \boxed{\alpha \beta = \beta \alpha}$$

Commuting

Algebraic Topology: a beginner's course

next in the series:

AlgTop26: Covering spaces